EXP:21

CODE:

```c
#include <stdio.h>

#define MAX_MEMORY 1000

int memory[MAX_MEMORY];

// Initialize memory
void initializeMemory() {
   for (int i = 0; i < MAX_MEMORY; i++) {
      memory[i] = -1;
   }
}

// Display memory
void displayMemory() {
   int i = 0;

   printf("\nMemory Status:\n");

   while (i < MAX_MEMORY) {
      if (memory[i] == -1) {
         int start = i;

         while (i < MAX_MEMORY && memory[i] == -1) {
            i++;
         }

         printf("Free block: %d - %d\n", start, i - 1);
      } else {
         i++;
      }
   }
}

// ✅ WORST FIT allocation
void allocateMemory(int processId, int size) {
   int maxBlockSize = 0;
   int maxStart = -1;

   int i = 0;

   while (i < MAX_MEMORY) {
      if (memory[i] == -1) {
         int start = i;
         int blockSize = 0;

         while (i < MAX_MEMORY && memory[i] == -1) {
            blockSize++;
            i++;
         }


}
```

```c
// Find largest block
         if (blockSize > maxBlockSize) {
            maxBlockSize = blockSize;
            maxStart = start;
         }
      } else {
         i++;
      }
   }

   // Allocate if enough space
   if (maxBlockSize >= size) {
      for (int i = maxStart; i < maxStart + size; i++) {
         memory[i] = processId;
      }

      printf("Allocated %d-%d to Process %d (Worst Fit)\n",
            maxStart, maxStart + size - 1, processId);
   } else {
      printf("Allocation failed for Process %d\n", processId);
   }
}

// Deallocate
void deallocateMemory(int processId) {
   for (int i = 0; i < MAX_MEMORY; i++) {
      if (memory[i] == processId) {
         memory[i] = -1;
      }
   }

   printf("Memory released by Process %d\n", processId);
}

int main() {
   initializeMemory();
   displayMemory();

   allocateMemory(1, 200);
   displayMemory();

   allocateMemory(2, 300);
   displayMemory();

   deallocateMemory(1);
   displayMemory();

   allocateMemory(3, 400);
   displayMemory();

   return 0;
}
```

OUTPUT:

C:\Users\Dell\Downloads\Unt

```
Process returned 32765 (0x7FFD)   execution time : 0.890 s
Press any key to continue.
```